RAG Architecture v1

1. Milvus Retrieval Layer

Milvus is the retrieval layer for the enterprise Agent Chat demo.
It stores text_vector for semantic search.
It stores sparse_vector for BM25 keyword retrieval.
It stores nullable image_vector for diagram evidence.
Metadata fields include source_type, doc_type, department, updated_at, priority, section, page_no, and chunk_id.

2. S3 Sync Flow

The S3 sync job performs bucket scanning.
It performs change detection for updated objects.
It extracts metadata from object storage.
It runs document parsing for Markdown, PDF pages, and images.
It performs chunking with stable doc_id and chunk_id values.
It runs embedding generation for text and optional images.
It performs Milvus insertion into the kb_chunks collection.
These facts support the S3 sync pipeline question.

3. Hybrid Retrieval

Hybrid retrieval combines dense vector search and sparse BM25.
Milvus applies metadata filter by source_type and department.
Search exposes order_by with updated_at desc and priority desc.
The UI shows aggregation by source_type, doc_type, department, and has_image_vector.

4. Agentic RAG Workflow

LangGraph runs classify_query and rewrite_query.
It then runs milvus_hybrid_retrieve and rerank_evidence.
grade_evidence decides whether to retry up to 3 times.
generate_answer_streaming returns cited answer chunks.